

Project Design Phase

Solution Architecture

Date	17 July 2026
Team ID	
Project Name	SB Stocks — MERN Stack Stock Trading Simulation Platform
Maximum Marks	4 Marks

Solution architecture is a complex process — with many sub-processes — that bridges the gap between the business problem and the technology solution. Its goals are to:

- Find the best tech solution to solve an existing business problem.
- Describe the structure, characteristics, behaviour, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

SB Stocks — Solution Architecture Diagram

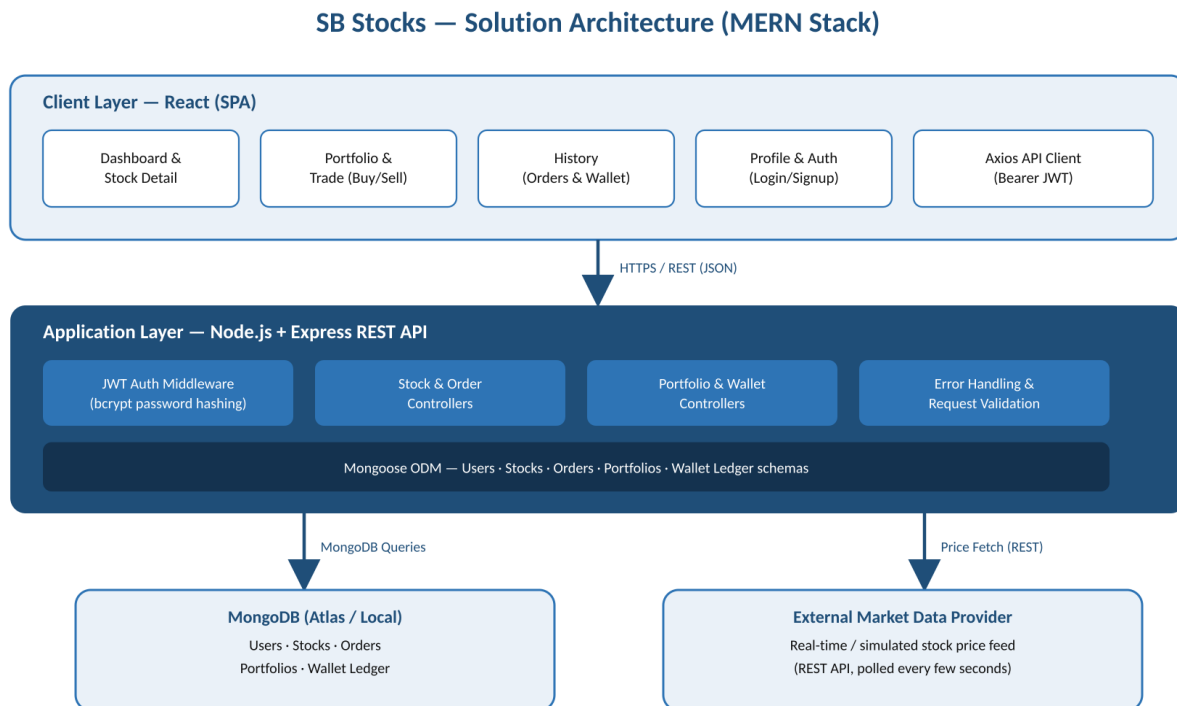


Figure 1: End-to-end architecture and data flow of SB Stocks — React client, Express/Node API, MongoDB, and the external market-data provider.

Component Overview

Layer	Technology	Responsibility
Client Layer	React (SPA) + Axios	Renders Dashboard, Stock Detail, Portfolio, History, and Profile pages; manages session state; calls the REST API with a Bearer JWT.
Application Layer	Node.js + Express.js	Exposes REST endpoints; JWT auth & bcrypt password hashing middleware; Stock, Order, Portfolio, and Wallet controllers; centralized error handling.
Data Layer	MongoDB + Mongoose	Persists Users, Stocks, Orders, Portfolios, and Wallet Ledger documents; enforces schema-level validation.
External Integration	Market Data Provider API	Supplies real-time / simulated stock prices, polled by the backend and cached for the live ticker and charts.

Data & Request Flow

- **1.** The React client sends a login/signup request; the API verifies credentials (bcrypt) and returns a signed JWT.
- **2.** Subsequent requests (dashboard, trade, portfolio, history) attach the JWT as a Bearer token; Express middleware verifies it before reaching a controller.
- **3.** The backend polls the external market-data provider on an interval and serves the latest prices to the Dashboard and Stock Detail views.
- **4.** A Buy/Sell request updates the user's Orders and Portfolio documents in MongoDB, and is written to the Wallet Ledger for full auditability.
- **5.** History and Portfolio pages read directly from MongoDB via Mongoose to render up-to-date holdings, returns, and trade logs.